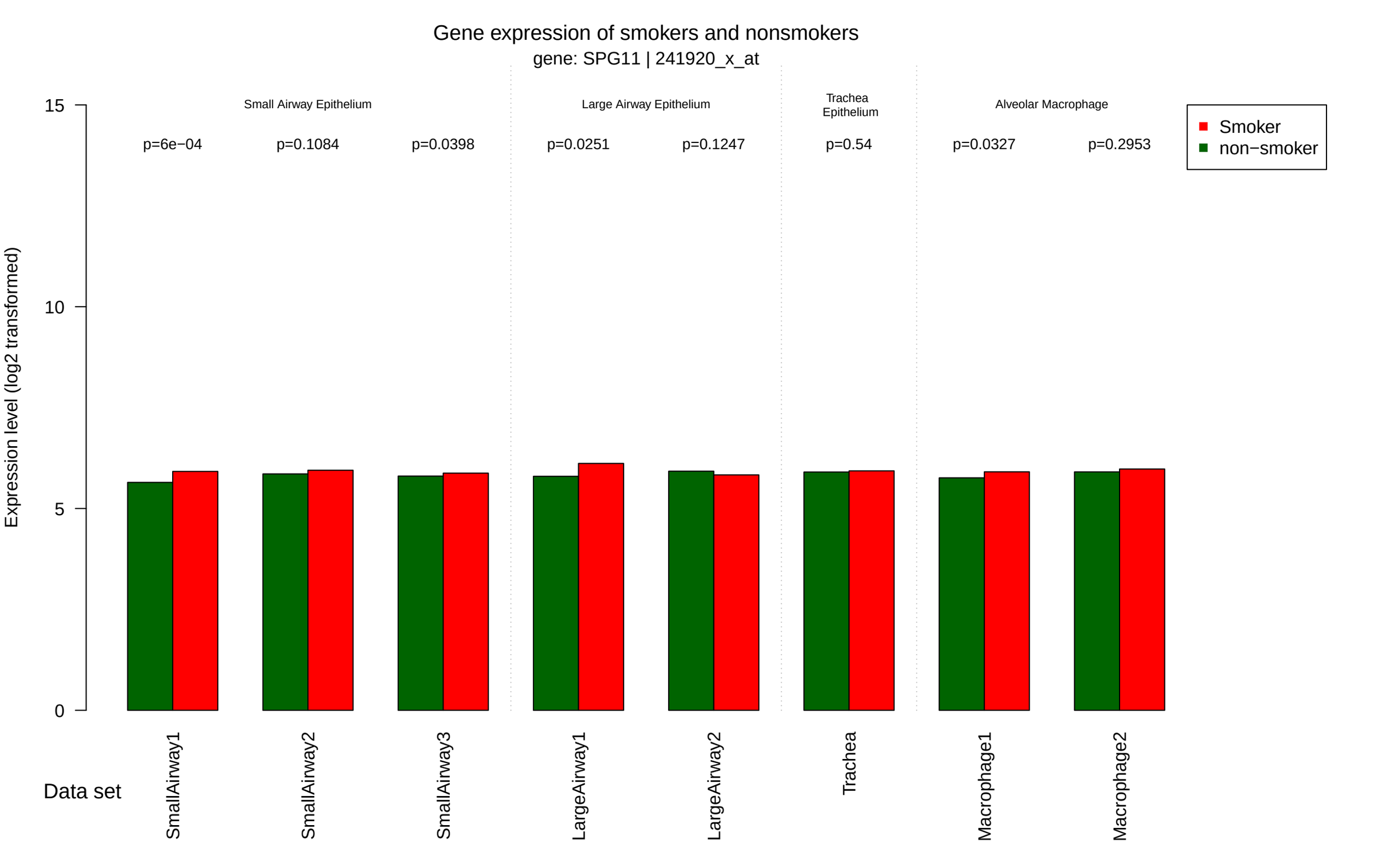
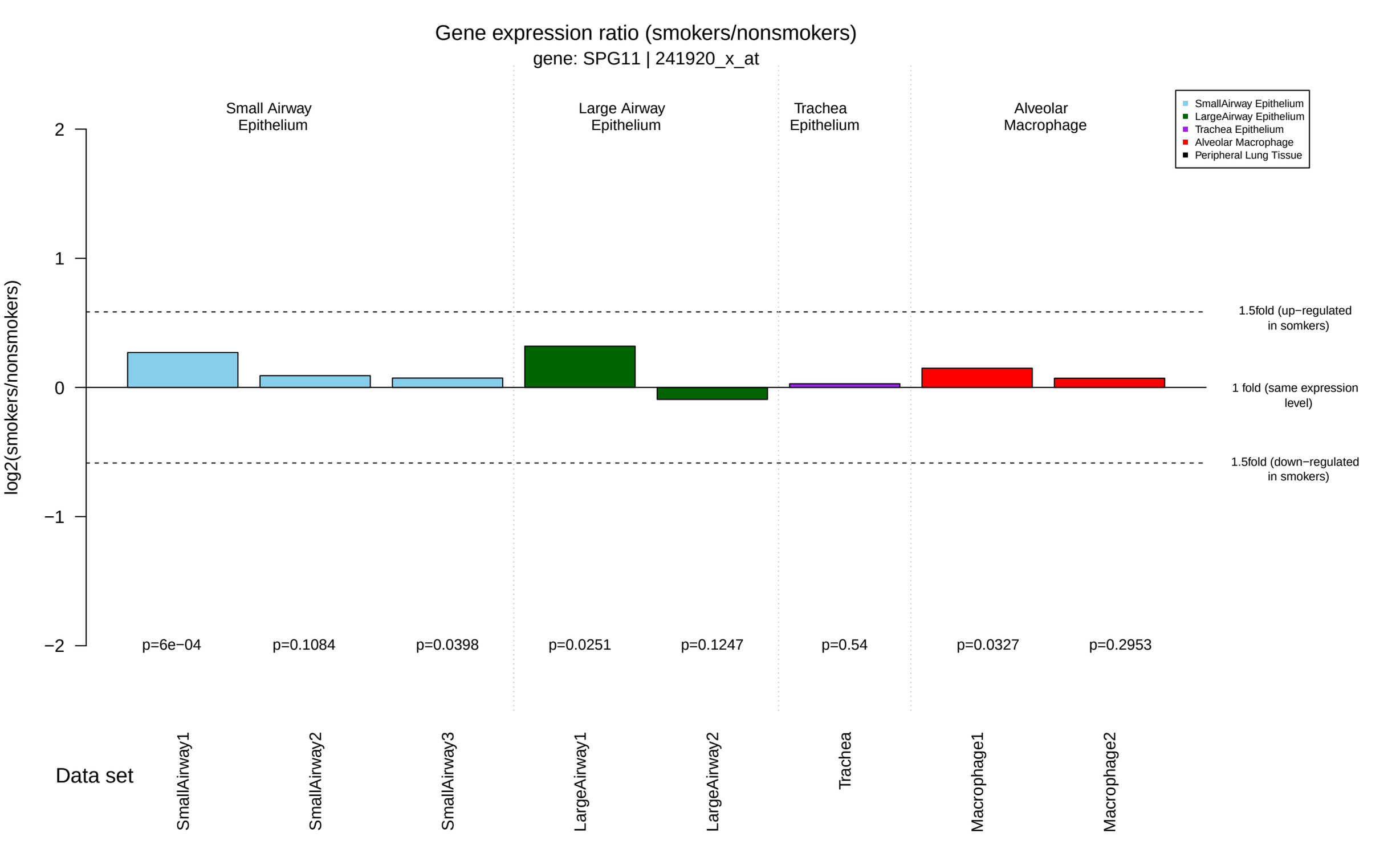
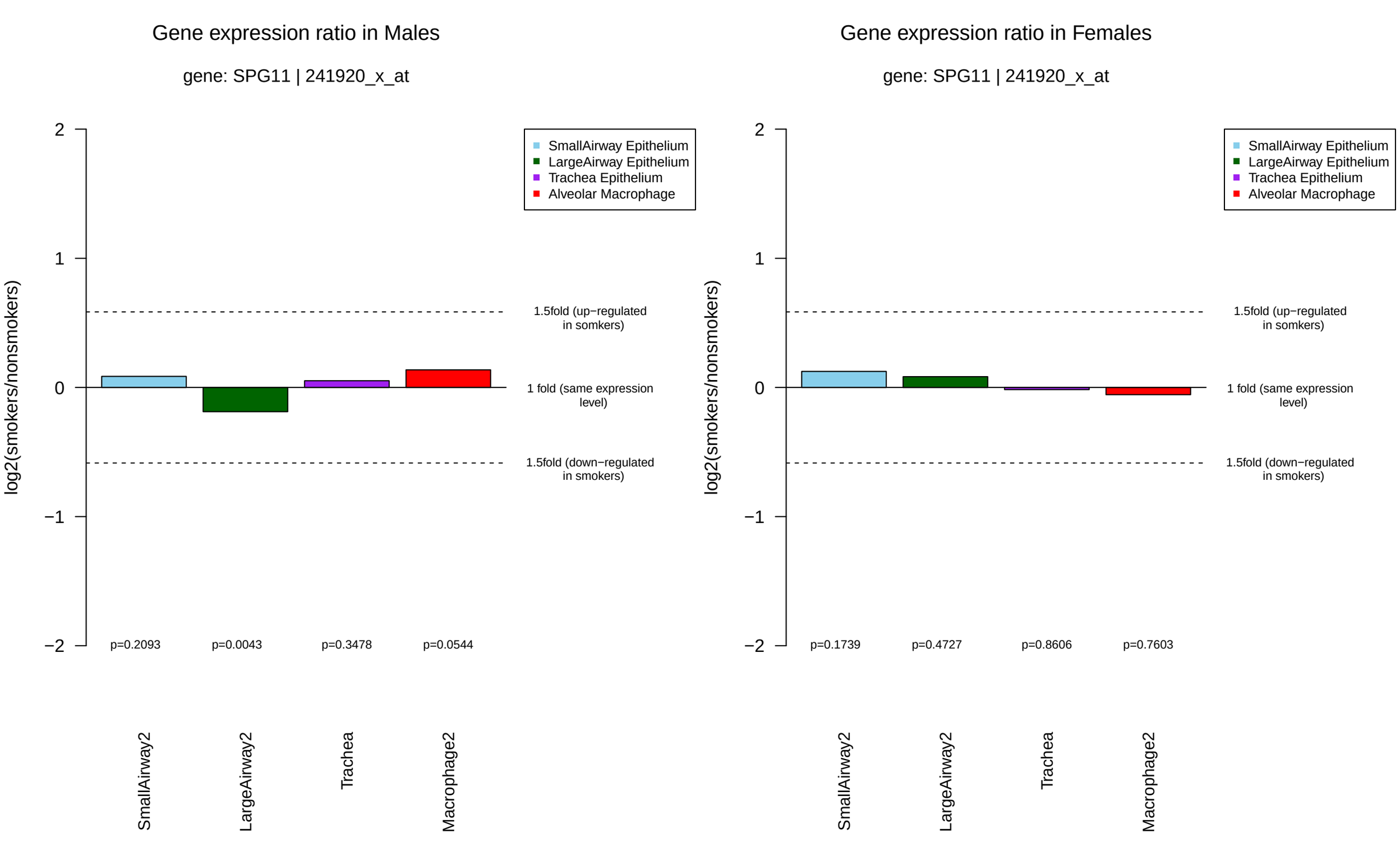
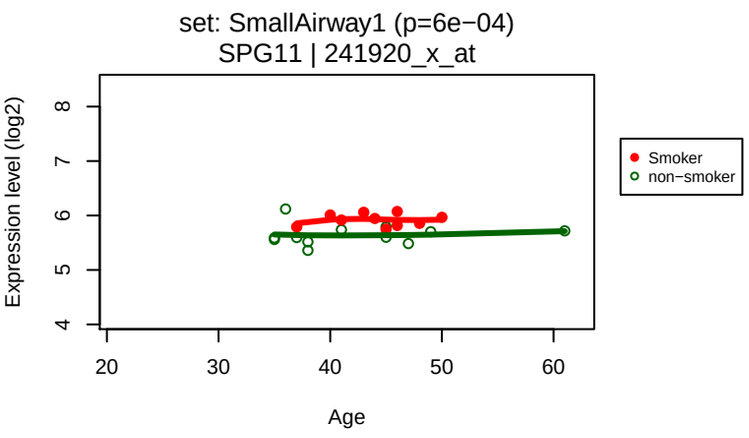
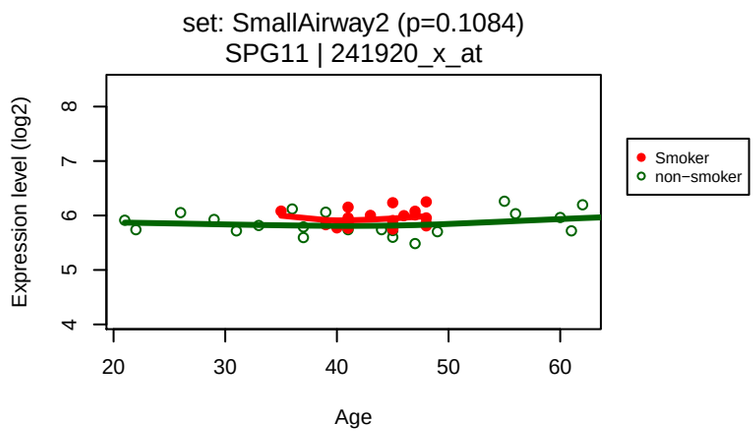
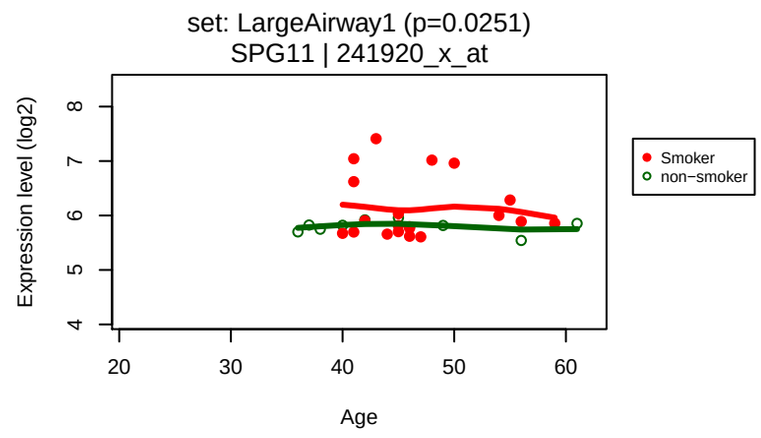

Gene expression of smokers and nonsmokers
gene: SPG11 | 241920_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=6e−04
p=0.1084
p=0.0398
p=0.0251
p=0.1247
p=0.54
p=0.0327
p=0.2953
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

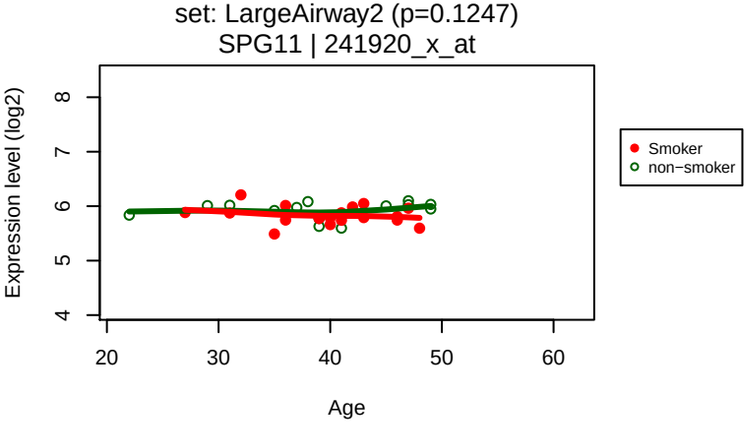

Gene expression ratio (smokers/nonsmokers)
gene: SPG11 | 241920_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.1084
p=0.0398
p=0.0251
p=0.1247
p=0.54
p=0.0327
p=0.2953
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

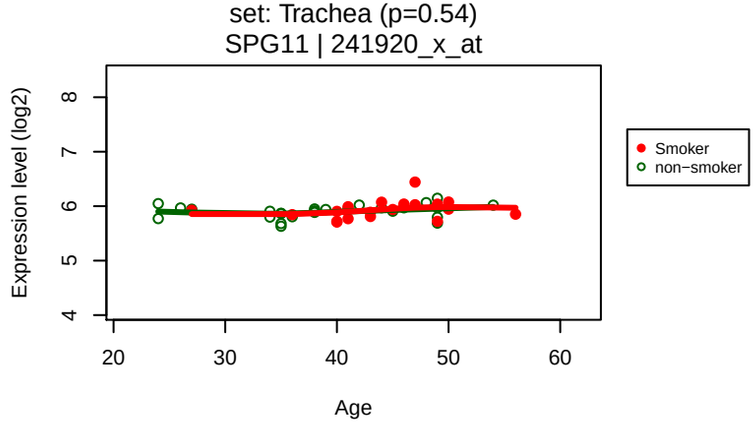

Gene expression ratio in Males
gene: SPG11 | 241920_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2093
p=0.0043
p=0.3478
p=0.0544
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPG11 | 241920_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1739
p=0.4727
p=0.8606
p=0.7603
SmallAirway2
LargeAirway2
Trachea
Macrophage2

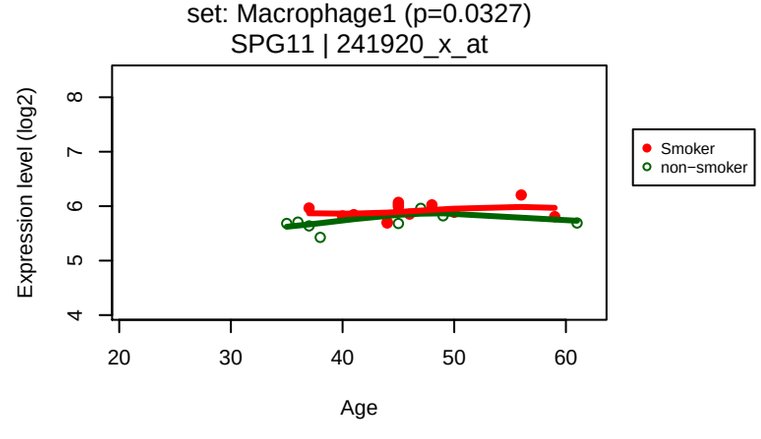

set: SmallAirway1 (p=6e−04)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker

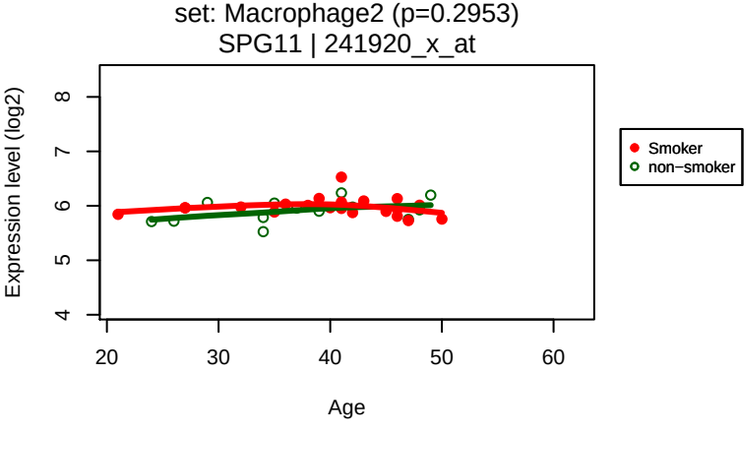

set: SmallAirway2 (p=0.1084)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker

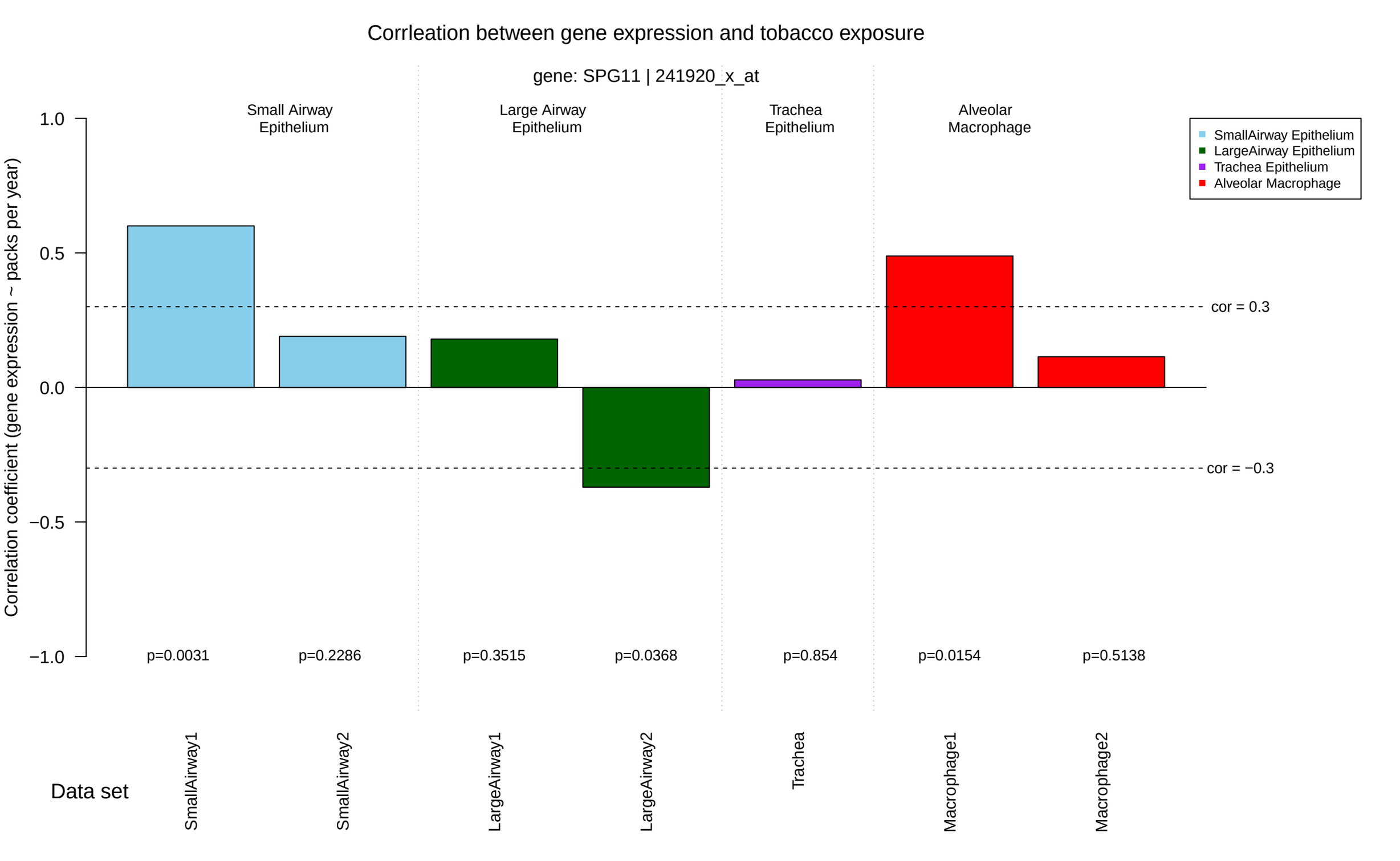

set: LargeAirway1 (p=0.0251)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1247)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.54)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0327)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2953)
SPG11 | 241920_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPG11 | 241920_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0031
p=0.2286
p=0.3515
p=0.0368
p=0.854
p=0.0154
p=0.5138
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2